Project: Olympic Athletes Performance Visual Analytics System.

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Milestone 1

Dataset Description:
https://www.kaggle.com/datasets/eshummalik/global-olympic-athletes-performance-dataset
We were able to source this dataset from kaggle.The dataset contains detailed historical records of Olympic athletes including their demographics, participation details, sports categories, and medal achievements. It covers multiple Olympic years, cities, and both Summer & Winter seasons.

The dataset contains 15 columns and 70,000 rows. Some columns describe the athletes' features (Sex, Age, Height, Weight, Gender, Name), while some give a description of the events they took part in (Sport, Event, Medal, City, Year, Season,Games.)

Column	Raw Type	Description
ID	Integer	Unique athlete identifier
Name	String	Full athlete name
Sex	String (M/F)	Athlete sex

Age	Float	Age at time of Games — 3.9% missing
Height	Float	Height in centimetres — 23.2% missing
Weight	Float	Weight in kilograms — 24.4% missing
Team	String	Country/team name
NOC	String	3-letter IOC country code
Games	String	Combined year + season label (e.g. '2012 Summer')
Year	Integer	Olympic year
Season	String	Summer or Winter
City	String	Host city
Sport	String	Sport category
Event	String	Specific event within sport
Medal	String	Gold / Silver / Bronze — 86.2% NaN (structural)

Data Assessment:

-The missing values in the medals column(60,310) was replaced with “No Medal” since those athletes participated but did not win any medal.Other missing values were for height (16,254),age (2,732) and weight (17,101).
-The missing values for age contributed 3.90% of the age values.
-The missing values for height contributed 23.22% of the height values.
-The missing values for weight contributed 24.43% of the weight values.

Descriptive Statistics:

-The mean age was 25.6 having the minimum age as 11 and the maximum age as 88.
-The average weight was 175.5 and the average height was 70.9.
-60,310 contestants did not win any medals, 3,292 won Gold medals , 3,208 won Bronze medals while 3,188 won Silver medals.

```
[134]: dataset = pd.read_csv("Olympics_dataset.csv")
       dataset.head(5)
```

[134]:

	ID	Name	Sex	Age	Height	Weight	Team	NOC	Games	Year	Season	City	Sport	Event	Medal
0	1	A Dijiang	M	24.0	180.0	80.0	China	CHN	1992 Summer	1992	Summer	Barcelona	Basketball	Basketball Men's Basketball	NaN
1	2	A Lamusi	M	23.0	170.0	60.0	China	CHN	2012 Summer	2012	Summer	London	Judo	Judo Men's Extra-Lightweight	NaN
2	3	Gunnar Nielsen Aaby	M	24.0	NaN	NaN	Denmark	DEN	1920 Summer	1920	Summer	Antwerpen	Football	Football Men's Football	NaN
3	4	Edgar Lindenau Aabye	M	34.0	NaN	NaN	Denmark/Sweden	DEN	1900 Summer	1900	Summer	Paris	Tug-Of-War	Tug-Of-War Men's Tug-Of-War	Gold
4	5	Christine Jacoba Aaftink	F	21.0	185.0	82.0	Netherlands	NED	1988 Winter	1988	Winter	Calgary	Speed Skating	Speed Skating Women's 500 metres	NaN

```
[142]: dataset.describe()
```

[142]:

	ID	Age	Height	Weight	Year
count	70000.000000	67268.000000	53746.000000	52899.000000	70000.000000
mean	18081.846986	25.644645	175.505303	70.900216	1977.766457
std	10235.613253	6.485239	10.384203	14.217489	30.103306
min	1.000000	11.000000	127.000000	25.000000	1896.000000
25%	9325.750000	21.000000	168.000000	61.000000	1960.000000
50%	18032.000000	25.000000	175.000000	70.000000	1984.000000
75%	26978.000000	28.000000	183.000000	79.000000	2002.000000
max	35658.000000	88.000000	223.000000	214.000000	2016.000000

INITIAL INSIGHTS:

-Most Olympic participants did not win medals.
-Athlete ages are concentrated around young adulthood.
-There is large missing data in height and weight fields, which requires cleaning in the next milestone.
-The dataset is rich enough for trend analysis, country comparisons, athlete profiling, and medal performance visualization.

MILESTONE 2

Data Cleaning

Handling Missing Values

Different strategies were applied based on data context:

- Medal column-Missing values replaced with "`No medal`" since missing implies no medal won.
- Age-Filled using median since it is more robust to outliers than mean
- Height & Weight-Filled using median grouped by Sport

Remaining missing values handled using global median.

Duplicate Removal

- Duplicate records were removed to improve data integrity.

Data Transformation

Aggregation

1. Top Countries by Medal Count
2. Medals per Country per Year-Created a structured dataset showing medal trends over time
3. Average Age per Sport-Helps compare athlete demographics across sports

Joining

Aggregated medal counts were merged back into the main dataset.

Feature Engineering

Several new variables were created to enhance analysis:

1. BMI (Body Mass Index)

Weight / (Height/100)^2-Provides insight into athlete body composition

2. Age Group

- Categorized athletes into:

    - Junior
    - Young Adult
    - Prime
    - Experienced
    - Veteran

Converts continuous data into meaningful categories.

3. Medal Winner Flag

- Binary variable:
    - 1 → Won a medal
    - 0 → Did not win

Useful for future predictive modeling.

4. Olympic Era

- Athletes grouped into historical periods:
    - Early
    - Post-War
    - Cold War
    - Modern
    - Contemporary

Enables time-based analysis.

<u>Data Quality Improvements</u>

After processing and creating new features, the overall quality of the dataset was greatly enhanced, making it more reliable and suitable for analysis.

1. Reduction of Missing Values

During preprocessing, missing or incomplete entries were handled appropriately.
This ensured that:

- More records could be used in analysis
- Bias caused by incomplete data was minimized

As a result, the dataset became more complete and dependable.

2. Improved Consistency

Data was standardized into uniform formats and categories.
For example:

- Age was grouped into defined categories (e.g., Prime, Veteran)
- Olympic periods were clearly labeled

This eliminated confusion and made comparisons across data easier and more meaningful.

3. Enhanced Interpretability

The introduction of new variables such as **BMI**, **Age Group**, and **Medal Winner Flag** made the dataset easier to understand.

Instead of working with raw numbers:

- BMI gives insight into body composition
- Age groups simplify continuous age data
- Binary medal flag clearly shows performance outcome

These transformations help both analysts and non-technical users quickly grasp insights.

4. Removal of Duplicate Records

Duplicate entries were identified and removed to ensure each observation is unique.

This improvement:

- Prevents overcounting or biased results
- Maintains data integrity
- Ensures accurate statistical analysis

After processing:

- Missing values significantly reduced.
- Dataset made more consistent and analysis-ready.
- New features improved interpretability.

- Duplicate records eliminated.